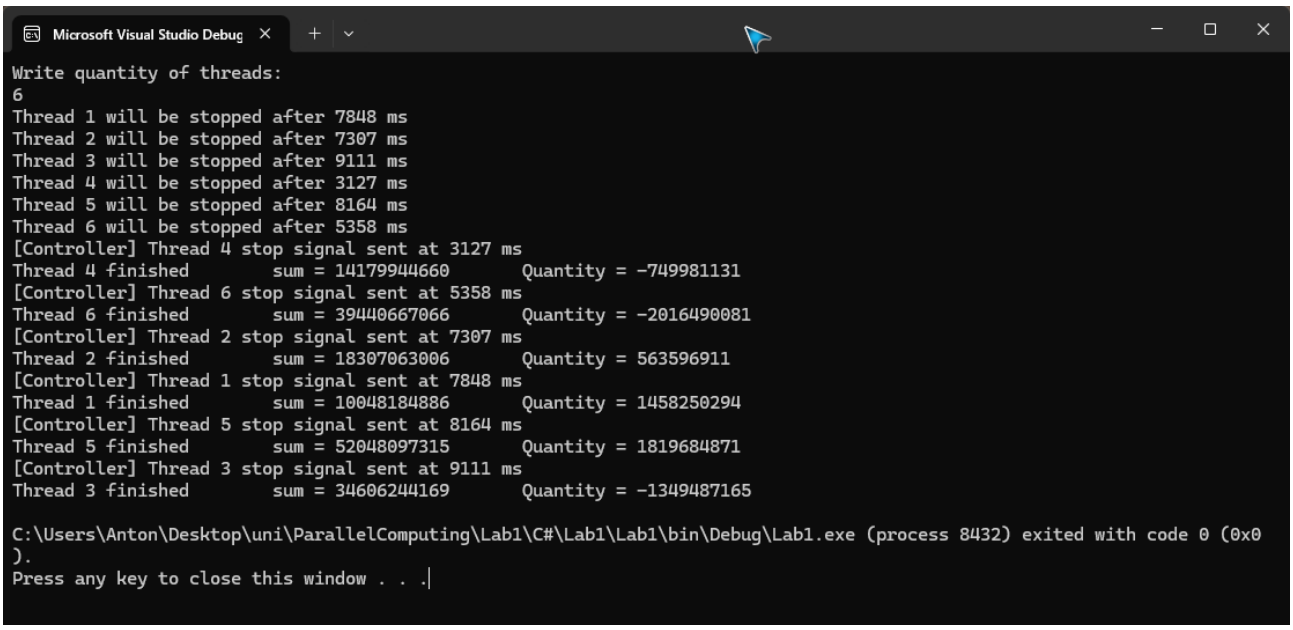
Лабораторна робота №1

Тема: Засоби створення багатопоточних програм.

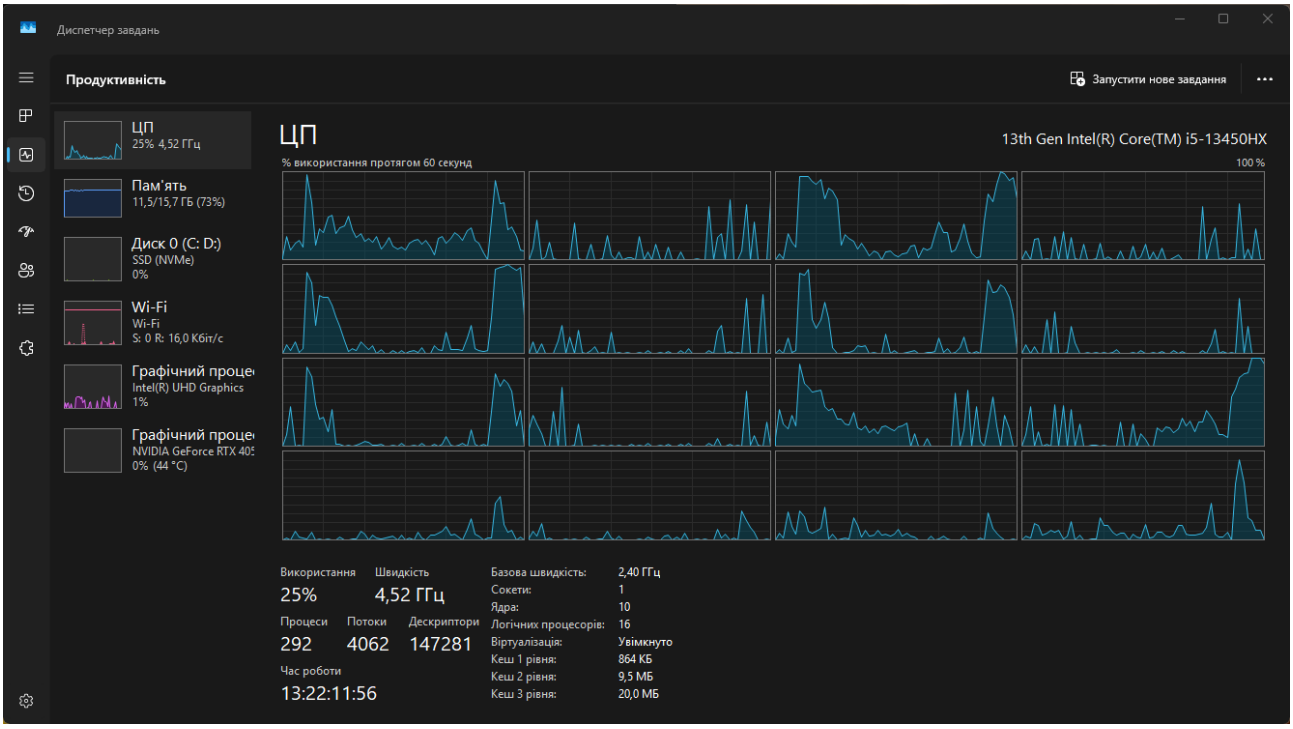
Мета: Познайомитись із засобами побудови паралельних програм різними мовами програмування.

Хід роботи

Робота програми при залученні 6 потоків:



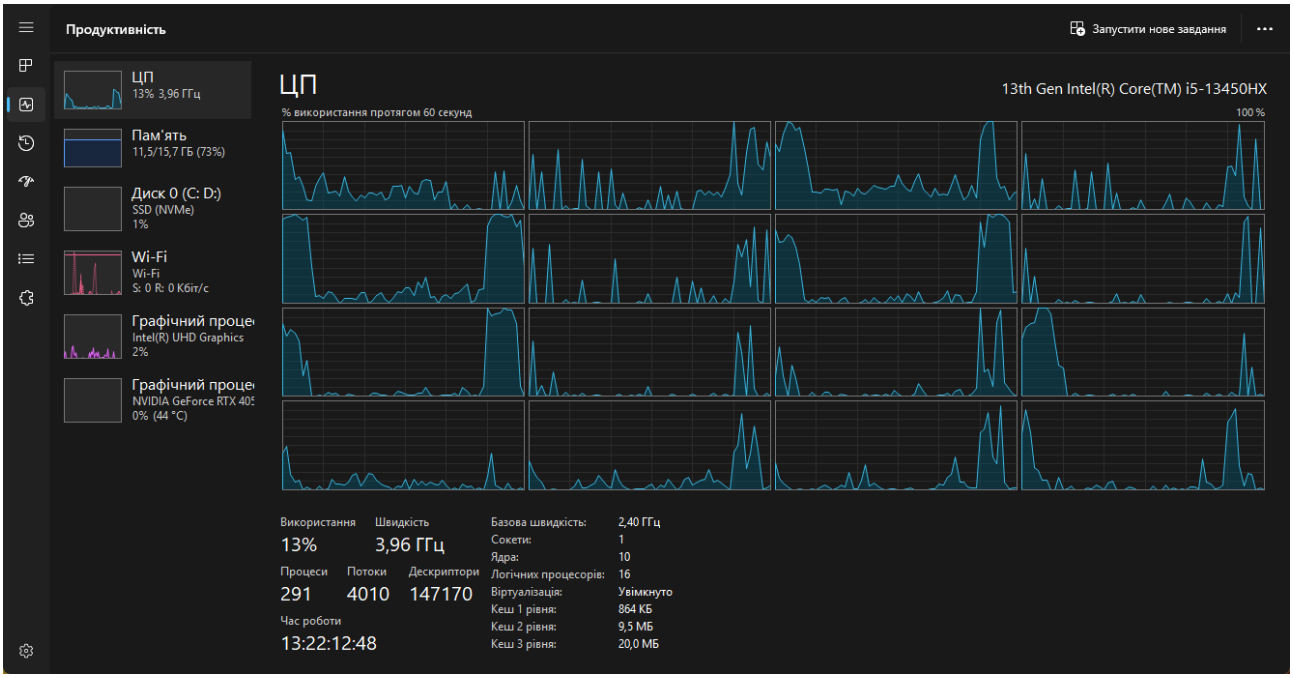
Навантаження 6 потоків:



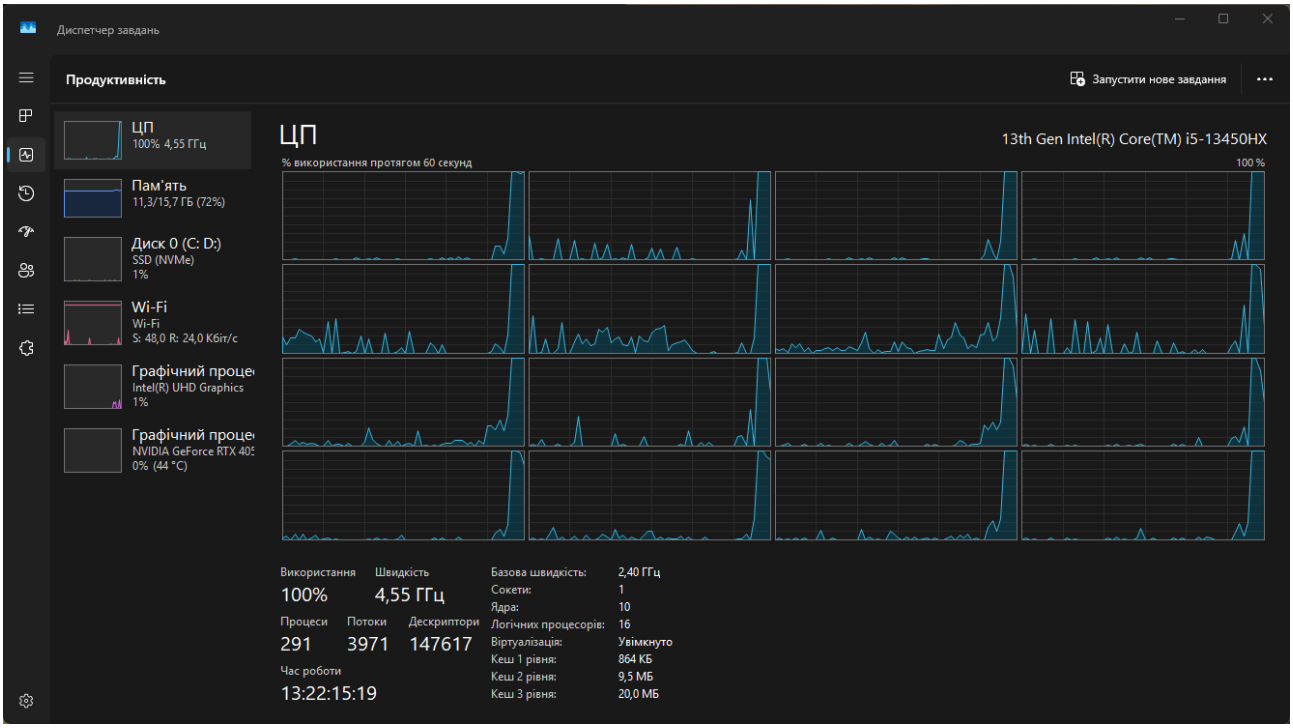
Робота програми при залученні 8 потоків:

```
Microsoft Visual Studio Debug

Write quantity of threads:
8
Thread 1 will be stopped after 7243 ms
Thread 2 will be stopped after 6103 ms
Thread 3 will be stopped after 9439 ms
Thread 4 will be stopped after 9817 ms
Thread 5 will be stopped after 6644 ms
Thread 6 will be stopped after 3486 ms
Thread 7 will be stopped after 7900 ms
Thread 8 will be stopped after 7092 ms
[Controller] Thread 6 stop signal sent at 3486 ms
Thread 6 finished        sum = 20283522558       Quantity = -914380203
[Controller] Thread 2 stop signal sent at 6103 ms
Thread 2 finished        sum = 12387720494       Quantity = 1898892951
[Controller] Thread 5 stop signal sent at 6644 ms
Thread 5 finished        sum = 36502644830       Quantity = -1289405626
[Controller] Thread 8 stop signal sent at 7092 ms
Thread 8 finished        sum = 54276192056       Quantity = -1805410585
[Controller] Thread 1 stop signal sent at 7243 ms
Thread 1 finished        sum = 7404542029        Quantity = -1185392563
[Controller] Thread 7 stop signal sent at 7900 ms
Thread 7 finished        sum = 55423085389       Quantity = -672350965
[Controller] Thread 3 stop signal sent at 9439 ms
Thread 3 finished        sum = 30123044007       Quantity = 1451080077
[Controller] Thread 4 stop signal sent at 9817 ms
Thread 4 finished        sum = 43924038280       Quantity = -1903892318

C:\Users\Anton\Desktop\uni\ParallelComputing\Lab1\C#\Lab1\Lab1\bin\Debug\Lab1.exe (process 14288) exited with code 0 (0x
0).
Press any key to close this window . . .
```

Навантаження 8 потоків:



Робота програми при залученні 8 потоків:

```
Microsoft Visual Studio Debug

Write quantity of threads:
16
Thread 1 will be stopped after 8168 ms
Thread 2 will be stopped after 4538 ms
Thread 3 will be stopped after 5223 ms
Thread 4 will be stopped after 8620 ms
Thread 5 will be stopped after 4549 ms
Thread 6 will be stopped after 4126 ms
Thread 7 will be stopped after 7354 ms
Thread 8 will be stopped after 8598 ms
Thread 9 will be stopped after 3319 ms
Thread 10 will be stopped after 2770 ms
Thread 11 will be stopped after 5954 ms
Thread 12 will be stopped after 6973 ms
Thread 13 will be stopped after 2851 ms
Thread 14 will be stopped after 4732 ms
Thread 15 will be stopped after 3314 ms
Thread 16 will be stopped after 8077 ms
[Controller] Thread 10 stop signal sent at 2770 ms
Thread 10 finished        sum = 17983379120        Quantity = 1798337912
[Controller] Thread 13 stop signal sent at 2851 ms
Thread 13 finished        sum = 24071066110        Quantity = 1851620470
[Controller] Thread 15 stop signal sent at 3314 ms
Thread 15 finished        sum = 37240587870        Quantity = -1812261438
[Controller] Thread 9 stop signal sent at 3319 ms
Thread 9 finished         sum = 22260456450        Quantity = -1821583246
[Controller] Thread 6 stop signal sent at 4126 ms
Thread 6 finished         sum = 17001511404        Quantity = -1461382062
[Controller] Thread 2 stop signal sent at 4538 ms
Thread 2 finished         sum = 6178385858         Quantity = -1205774367
[Controller] Thread 5 stop signal sent at 4549 ms
Thread 5 finished         sum = 16045224560        Quantity = -1085922384
[Controller] Thread 14 stop signal sent at 4732 ms
```

Навантаження 16 потоків: